LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

# Lesson 9: Project: My Simulation

Module 3 | 50 minutes | Project: simulation | Grades 6–8 · ages 11–14

**Objective:** Plan, build and test a simulation of a real-world process, run it as an experiment and report what you find.

**Success criteria**

- My simulation has rules, randomness and a measurement.
- I ran an experiment by changing one setting.
- I can explain my model's assumptions.

## Key vocabulary

- **assumption**: Something the model treats as true or leaves out.
- **controlled experiment**: Changing one thing and comparing.
- **measurement**: A value recorded to answer the question.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

## Timed lesson plan (50 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Rumour game (5 min): one learner whispers to two others each "round"; count how many know after 3 rounds. |
| 5 min | **Watch & plan** | Play the project video; learners plan on the project brief. |
| 25 min | **Build** | Build-along steps, then Rule, assumption or measurement?. Check in halfway: everyone has something working. |
| 7 min | **Test & peer feedback** | Swap and test each other's work: two stars and a wish. |
| 8 min | **Improve & submit** | Improve one thing; Project: My Simulation. |

## Unplugged warm-up

Rumour game (5 min): one learner whispers to two others each "round"; count how many know after 3 rounds.

## Teaching the idea

Learners apply simulation: define a question, rules, assumptions and a measurement; implement with agents or time steps; run a controlled experiment (change one variable). Emphasise limits of models.

**Analogy:** A wind tunnel: test ideas safely before the real thing.

**Live demo:** Show the sample epidemic model; change the speed and compare infected counts.

## Questions to ask

| Ask | Listen for |
|---|---|
| What is your question? | Learner-specific. |
| What did you change in the experiment? | One setting. |
| What does your model leave out? | Assumptions. |

## Differentiation

**Support**
Remix the particles or rabbit model with one new rule.

**Stretch**
Record measurements over time into a list and report the trend.

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| One run proves the answer | Run several times; results vary. |
| Change many settings at once | Change one to see its effect. |

## Assessment

Project rubric (also on the project brief):

| Criterion | 1 · Beginning | 2 · Developing | 3 · Secure | 4 · Excellent |
|---|---|---|---|---|
| **Model design** | No rules | Vague rules | Clear rules and assumptions | Thoughtful rules with justified assumptions |
| **Implementation** | Broken | Partly works | Agents or steps with randomness work | Clean, well-structured simulation |

| | | | | |
|---|---|---|---|---|
| **Measurement** | None | Shown but unclear | Clear measurement reported | Measured over time or across runs |
| **Experiment** | None | Changed several things | Changed one thing and compared | Repeated runs and fair comparison |
| **Conclusion** | None | Describes output | Answers the question | Answers and evaluates the model’s limits |

**Exit ticket**
One assumption your model makes.

**Homework**
Explain your simulation to someone at home and ask what they think it leaves out.

# Answer key — Kitabash activities

### 1. Rule, assumption or measurement? (perform · categorize)

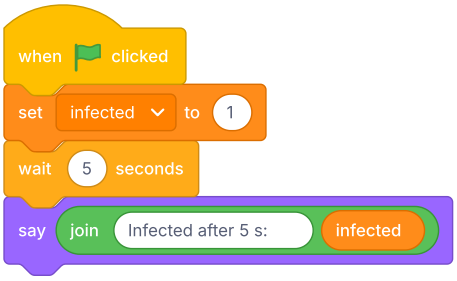
**Rule:** Each fish moves 3 steps in a random direction, If a shark touches a fish, the fish disappears
**Assumption:** The pond has no predators, All fish swim at the same speed
**Measurement:** Count fish at the end of each day

## 2. Project: My Simulation (probe · code lab)

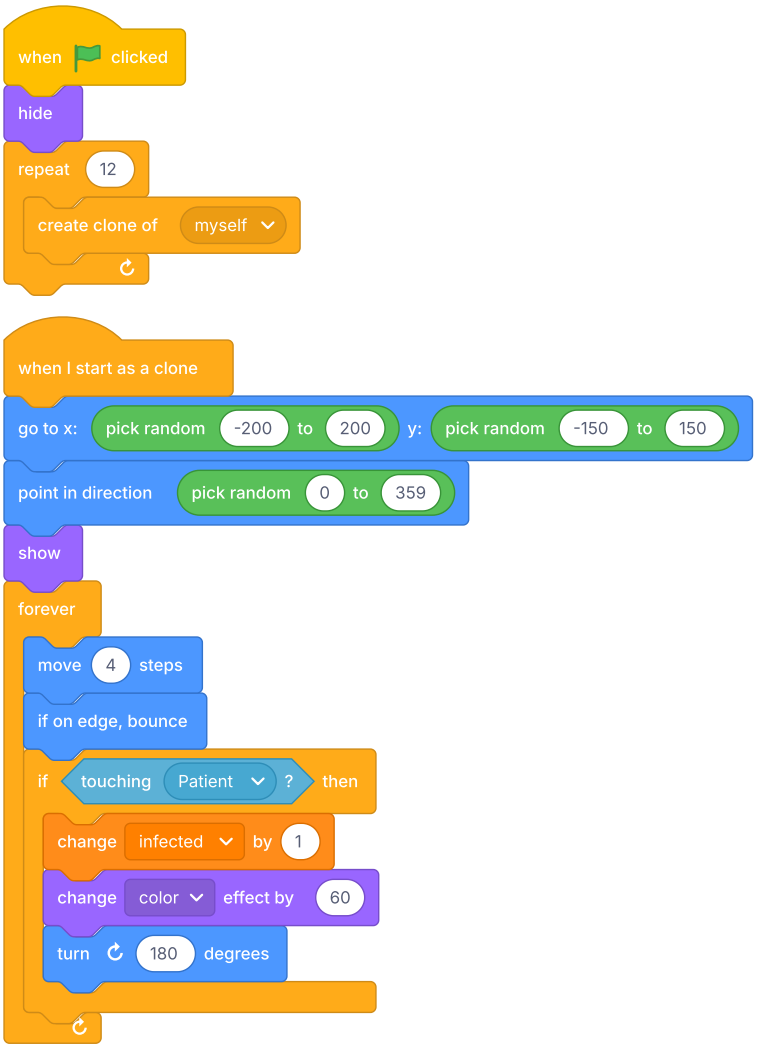
**Goals checked:** Randomness · Agents or time steps · Measures something · Reports a result

**Example project:**

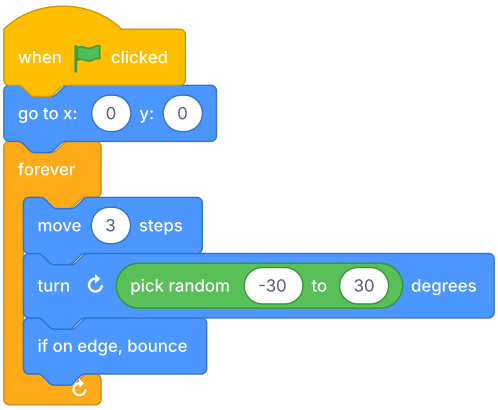
### Scientist




### Person




### Patient




## Answer key — worksheet

1. **Key words:** assumption = Something the model treats as true or leaves out. · controlled experiment = Changing one thing and comparing. · measurement = A value recorded to answer the question.
2. **Rule, assumption or measurement?:** Rule: Each fish moves 3 steps in a random direction, If a shark touches a fish, the fish disappears · Assumption: The pond has no predators, All fish swim at the same speed · Measurement: Count fish at the end of each day
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.